

Drug Abuse and Addiction

Chapter 9

Learning objectives – 3 key points

What is addiction?

How do we study addiction in animal models?

- How does addiction develop?
- What maintains addiction?

How do substances change the brain?

- Neurobiology of addiction
- Substance-induced neuroadaptations

What is addiction?

Addiction is a chronic, relapsing behavioral disorder

Behavioral

compulsive drug-seeking and drug-using behavior

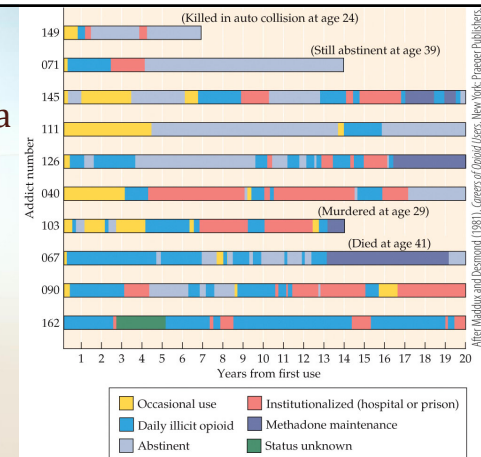
cravings - strong urges to take the drug

Chronic, relapsing

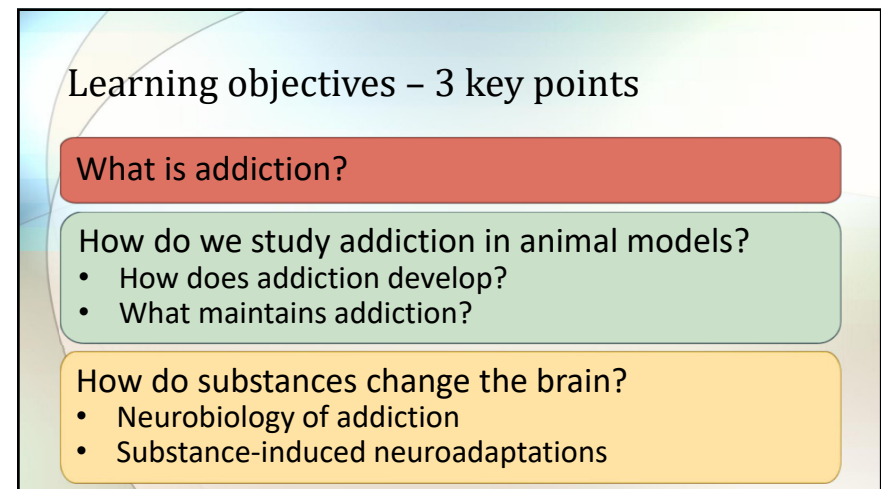
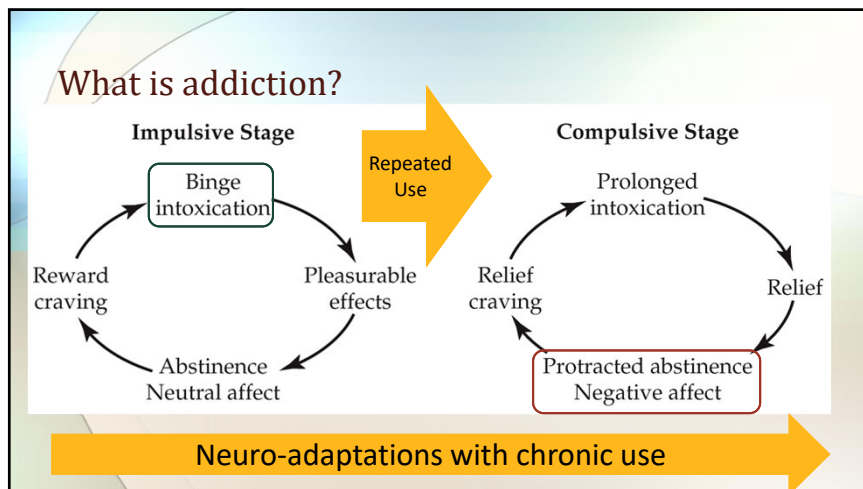
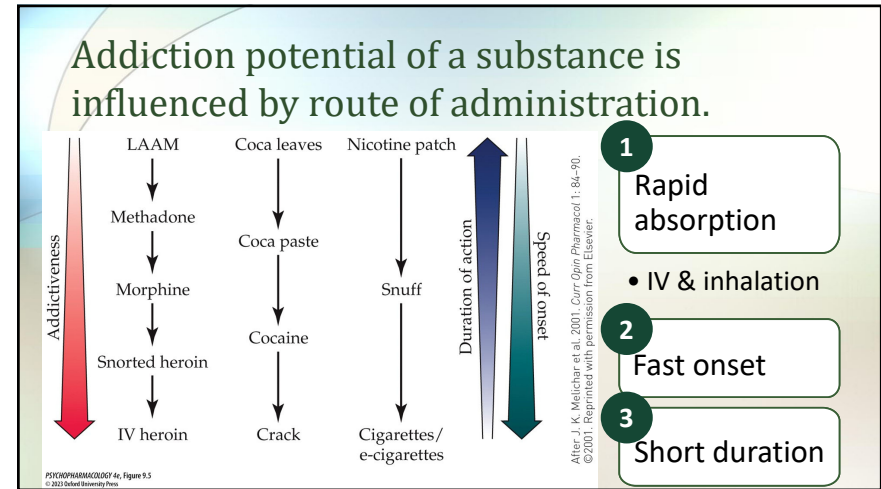
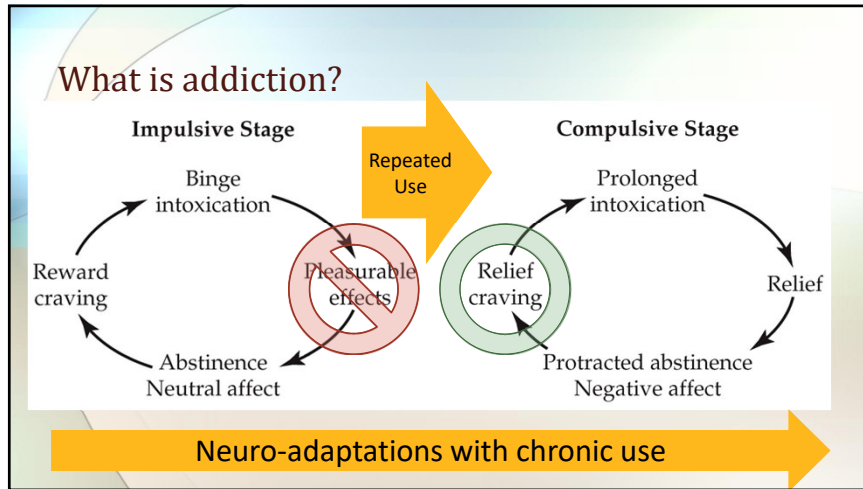
long-lasting

drug-free periods followed by relapse to drug use

Addiction is characterized by a chronic tendency to relapse



PSYCHOPHARMACOLOGY 3e, Figure 9.5
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Drugs of abuse share 2 important properties

Rewarding

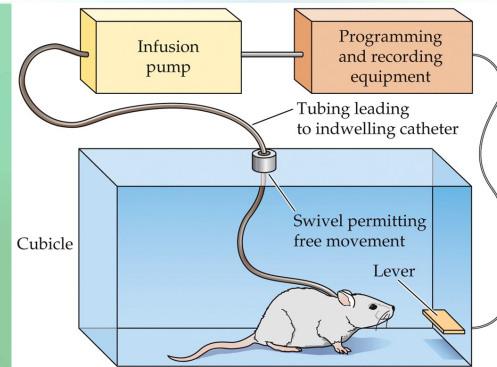
- the positive experience associated with the drug
- the “high”

Positively reinforcing

- consuming the drug strengthens whatever preceding behavior was performed
- how hard an animal will work for a drug

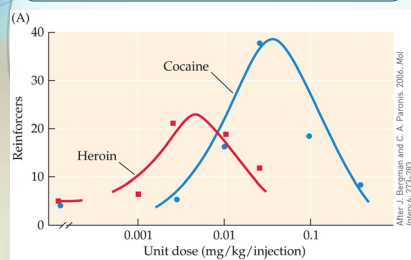
How hard will an animal work for drugs?

Drug self-administration

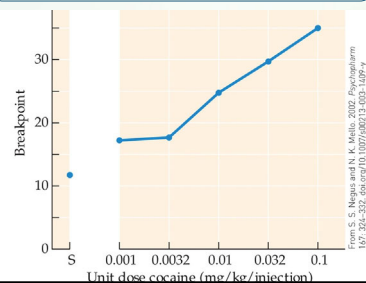


How hard will an animal work for drugs?

Dose-response

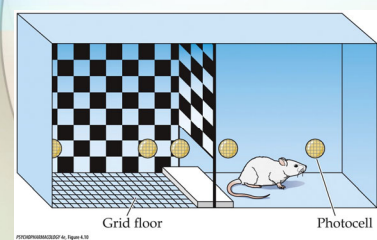


Breaking point

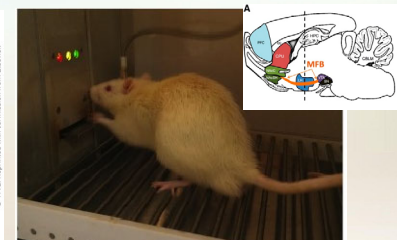


Rewarding properties of drugs are commonly evaluated using

Place conditioning



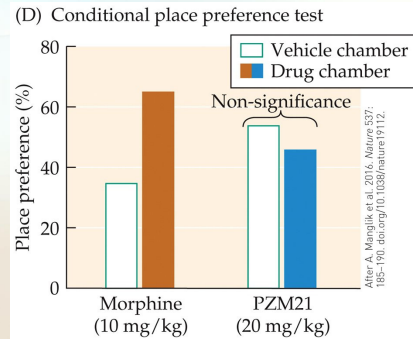
Electrical self-stimulation



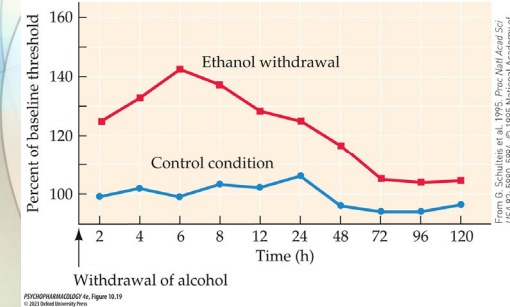
Negus & Mello, (2014). Intracranial self-stimulation to evaluate abuse potential of drugs. *Pharmacological Reviews*, 66(3), 869-917. <https://doi.org/10.1124/pr.112.007419>

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Animals conditioned to associate a particular chamber with morphine show preference for that chamber



Withdrawal from ethanol increases threshold for electrical self-stimulation of reward pathway



Reinforcement is less rewarding that it was during initial ethanol use

Relapse to drug use is modeled by reinstatement

1 Forced abstinence

- extinguishes drug seeking behavior (CR)

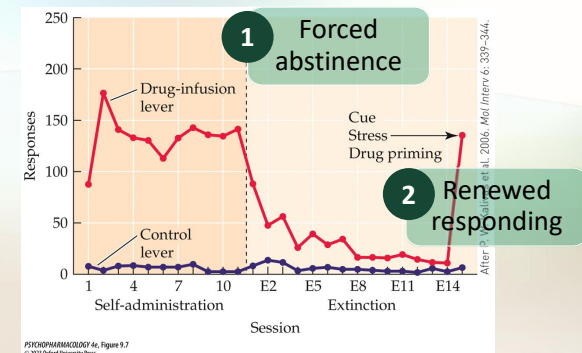
2 Renewed responding

- Some stimuli are effective in re-initiating drug-seeking = reinstatement

- (1) Drug priming (US)
- (2) Stress
- (3) CS's

CS's, stress, or drug exposure reinstates drug-seeking

Reinstatement is an animal model of relapse to drug use after abstinence



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Drug-taking behavior progresses from

 an **impulsive** stage

 to a **compulsive** stage

Koob, G. F. and Le Moal, M. (2005). Plasticity of reward neurocircuitry and the "dark side" of drug addiction. *Nature Neuroscience*, 8(11), 1442-1444. <https://doi.org/10.1038/nn1105-1442>

Drug-taking behavior progresses from

 an **impulsive** stage

- driven by positive reinforcement, drug reward properties
- activation of the reward circuitry
- generally occurs early in drug using behavior

 to a **compulsive** stage

Koob, G. F. and Le Moal, M. (2005). Plasticity of reward neurocircuitry and the "dark side" of drug addiction. *Nature Neuroscience*, 8(11), 1442-1444. <https://doi.org/10.1038/nn1105-1442>

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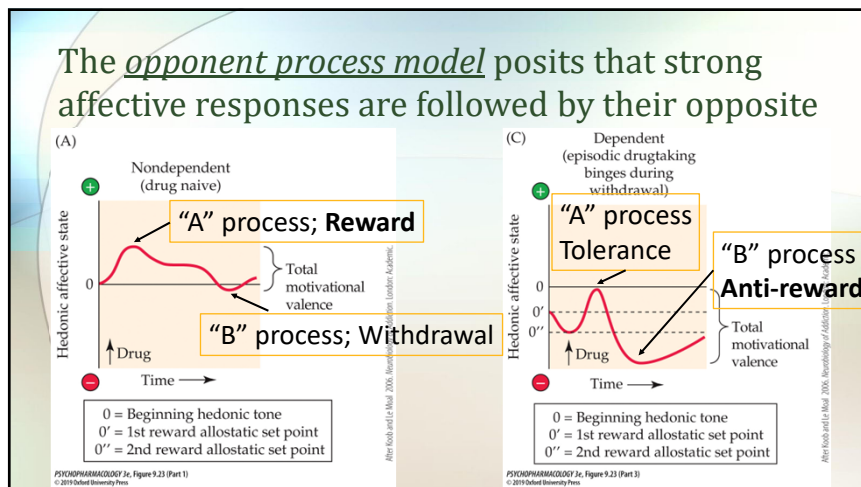
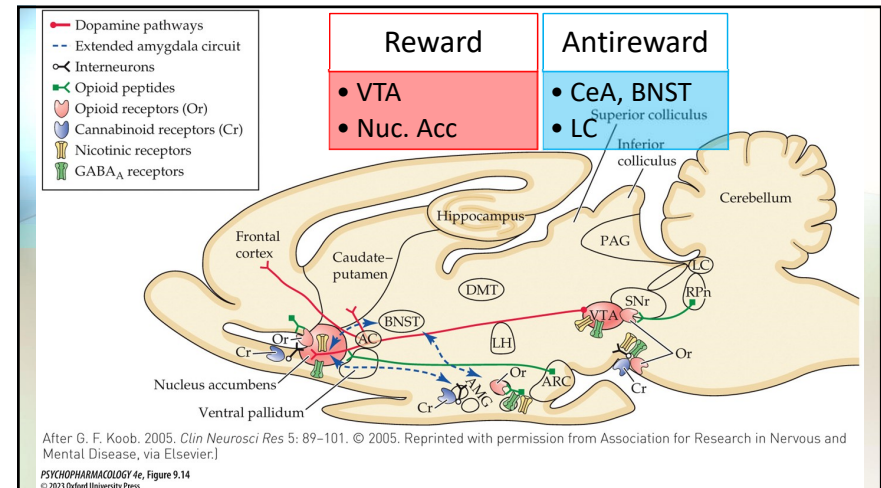
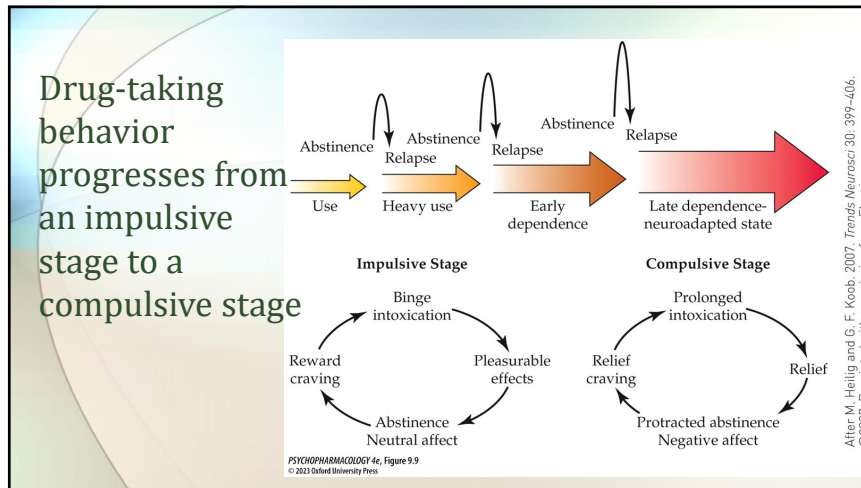
 an **impulsive** stage

 to a **compulsive** stage

- driven by negative reinforcement; removal of negative stimuli (withdrawal / abstinence)
- recruitment of anti-reward circuitry
- generally occurs later in drug using behavior

Koob, G. F. and Le Moal, M. (2005). Plasticity of reward neurocircuitry and the "dark side" of drug addiction. *Nature Neuroscience*, 8(11), 1442-1444. <https://doi.org/10.1038/nn1105-1442>

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DA is critical in reward

Drugs of abuse increase DA signaling in the mesolimbic reward pathway

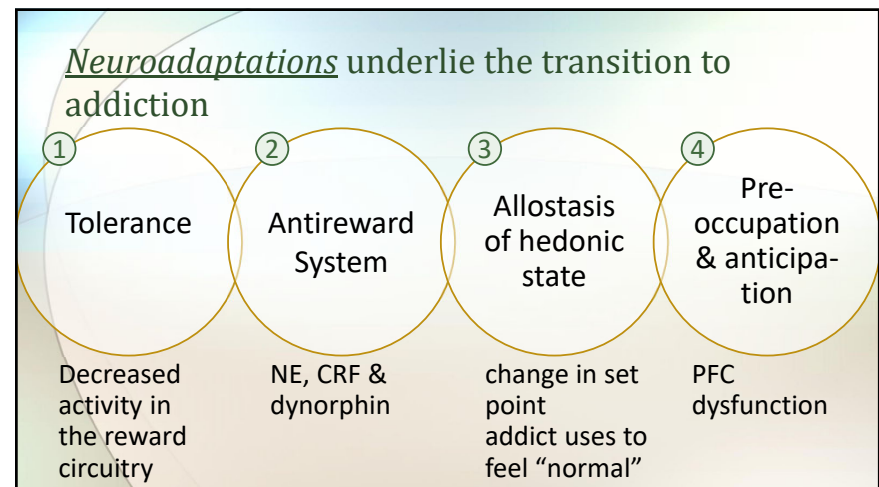
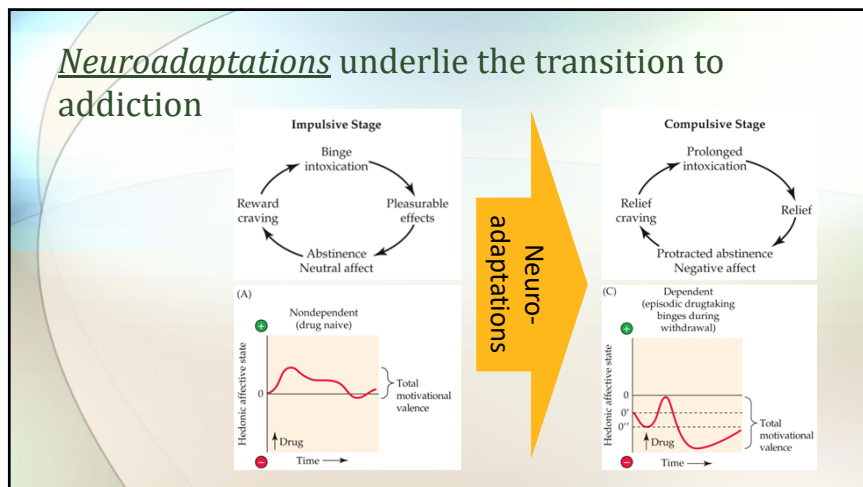
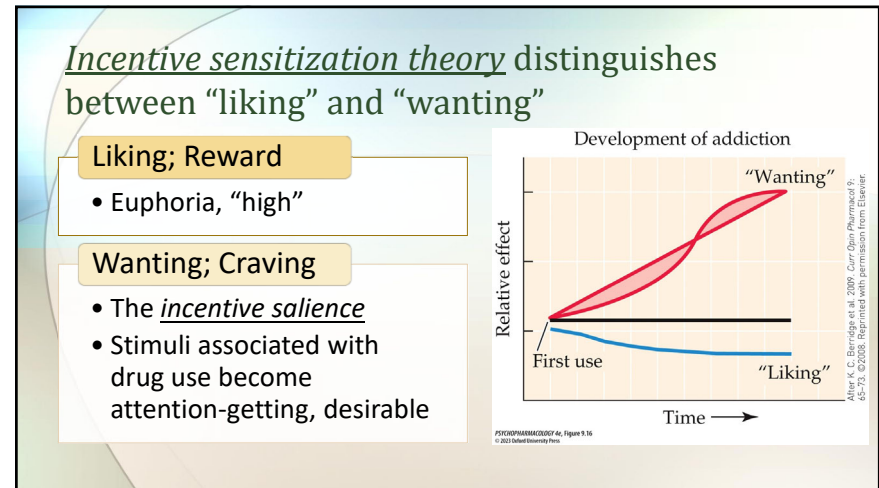
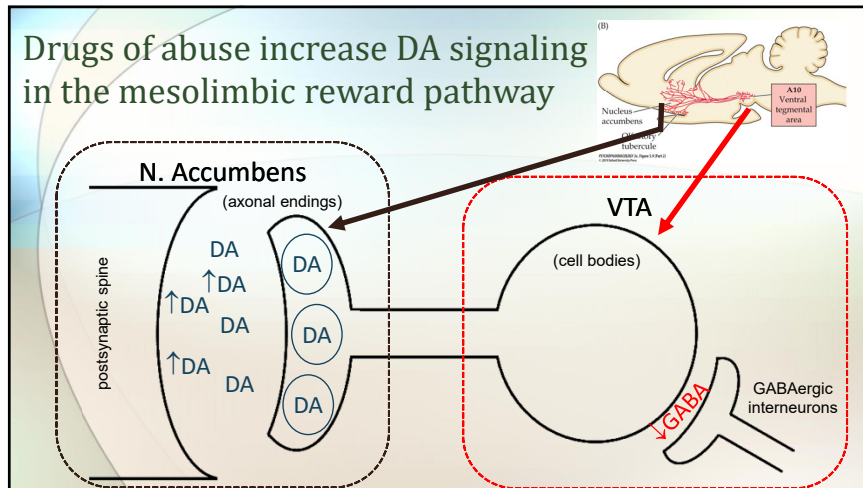
- DA-ergic transmission from VTA → N. accumbens
- mesocortical pathway VTA → PFC is also involved

Opioid, GABA, nACh receptors are also important

Incentive sensitization theory

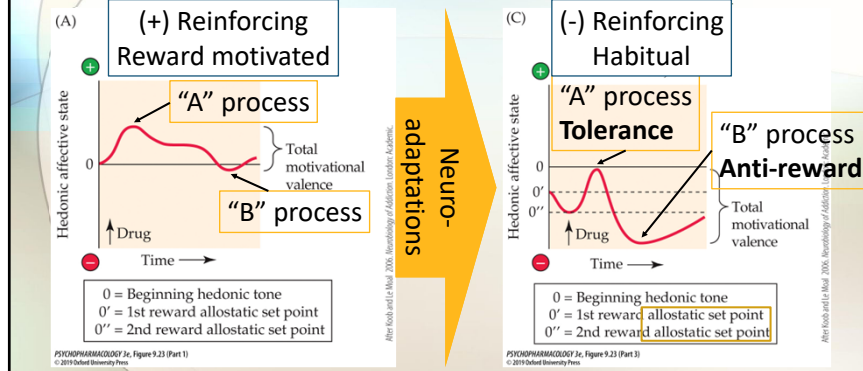
- Distinguishes between "liking" and "wanting"

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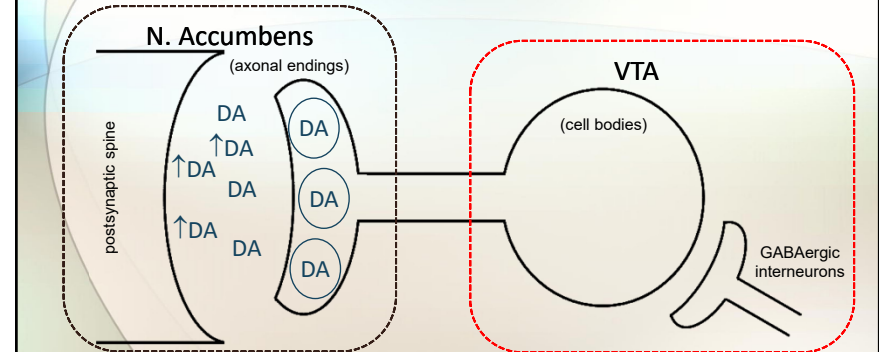


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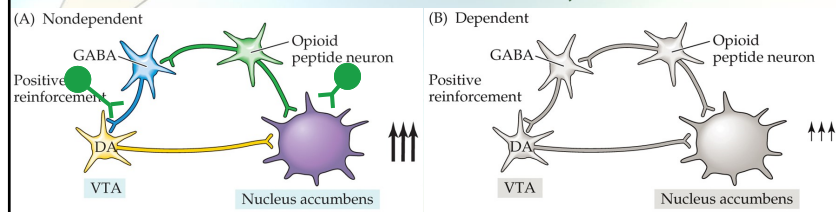
The opponent process model posits that strong affective responses are followed by their opposite



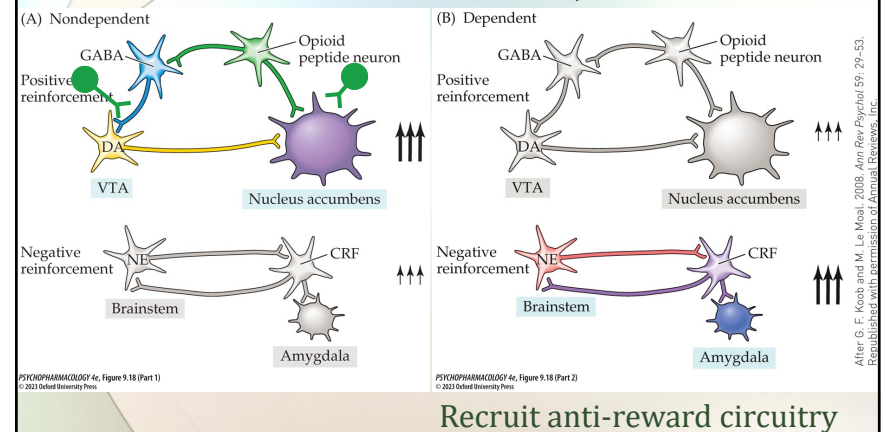
Diminished DA signaling is involved in tolerance and allostasis of hedonic set point



Tolerance; Allostasis



Tolerance; Allostasis



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Dysfunction in the PFC contributes to the preoccupation / anticipation phase

PFC is essential for behavioral control

- decision making, strategic planning; impulse control

PFC exerts “top-down” control over striatal systems

PFC dysfunction in addiction is characterized by

- intrusive thoughts
- craving (motivational valence)
- poor impulse control

The transcription factor $\Delta FosB$ is important in driving molecular changes underlying addiction

$\Delta FosB$ induction

$\Delta FosB$ -induced changes in gene expression

Epigenetic mechanisms

Summary: Neurobiology of Addiction

